PassFrame tuition

Speed, Distance & Time

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: A cyclist travels 63 km in 1 hour 45 minutes. What is her average speed?

Method & answer: 1 hour 45 minutes = 1.75 hours. 63 ÷ 1.75 = 36 km/h.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A car travels 315 km in 4.5 hours. What is its speed?

2. Which formula rearranges D = S × T to find speed? A) S = D × T B) S = D ÷ T C) S = T ÷ D D) S = T − D

3. A cyclist rides at 22 km/h for 3.5 hours. How far does he travel?

4. A train covers 434 km in 7 hours. What is its speed?

5. Which of these is a sensible average speed for a high-speed train? A) 8 km/h B) 45 km/h C) 250 km/h D) 2,500 km/h

6. A walker moves at 5.5 km/h for 2.4 hours. How far does she walk?

7. A car travels 279 km in 4.5 hours. What is its average speed?

8. A bus travels at 48 km/h for 3.5 hours. How far does it travel?

9. Select ALL the speed calculations below that are correct: A) 468 km in 6 hours = 78 km/h B) 245 km in 3.5 hours = 68 km/h C) 315 km in 5 hours = 63 km/h D) 176 km in 8 hours = 22 km/h E) 592 km in 8 hours = 76 km/h

10. A car travels 372 km in 6 hours. What is its speed?

11. A runner covers 16.8 km in 2.4 hours. What is her speed?

12. Which of these speeds is the fastest? A) 86 km/h B) 79 km/h C) 91 km/h D) 84 km/h

13. A coach travels at 72 km/h for 3.5 hours. How far does it travel?

14. A car takes 5.5 hours to travel 396 km. What is its average speed?

15. A cyclist travels 38 km in the first 2 hours and 42 km in the next 2 hours. What is her average speed over the whole journey?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A ship sails 483 km in 6.9 hours. What is its speed?

2. A car travels at 88 km/h for 2 hours 15 minutes. How far does it travel?

6. A cyclist travels at 30 km/h for 140 minutes. How far does she travel?

7. A lorry travels at 76 km/h for 4.5 hours. How far does it travel? A) 324 km B) 332 km C) 342 km D) 350 km

11. A train travels at 156 km/h. Select ALL statements that are TRUE: A) In 2.5 hours it covers 390 km B) In 30 minutes it covers 78 km C) In 20 minutes it covers 50 km D) In 1 hour 15 minutes it covers 195 km E) In 10 minutes it covers 26 km

12. A ferry takes 5.4 hours to travel 405 km. What is its average speed?

3. A train travels 522 km in 6 hours. What is its speed? A) 82 km/h B) 85 km/h C) 87 km/h D) 90 km/h

4. A car travels for 50 minutes at 108 km/h. How far does it travel?

5. A plane flies 4,080 km in 6 hours. What is its speed?

8. A car covers 555 km in 6 hours. What is its speed?

9. A train travels for 24 minutes at 165 km/h. How far does it travel?

10. A walker travels at 7.2 km/h for 1 hour 50 minutes. How far does he walk?

13. A coach travels at 84 km/h for 2 hours 50 minutes. How far does it travel?

14. Which journey has the greatest speed? A) 322 km in 3.5 hours B) 408 km in 4.8 hours C) 494 km in 5.2 hours D) 456 km in 6 hours

15. A car travels 180 km in the first 2 hours and 300 km in the next 3 hours. What is its average speed for the whole journey?

Standard / Advanced — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. A car's average speed for a 372 km journey is 62 km/h. How many minutes does the journey take?

2. A train travels 297 km in 3 hours 18 minutes. What is its average speed? A) 85 km/h B) 88 km/h C) 90 km/h D) 92 km/h

3. A plane's speed is 780 km/h. How far does it travel in 50 minutes?

4. A cyclist travels at 44 km/h. How long, in minutes, does she take to travel 11 km?

5. A car travels 375 km at an average speed that is 20 km/h faster than 55 km/h. How long does the journey take? A) 4.5 hours B) 5 hours C) 5.5 hours D) 6 hours

6. A lorry travels the first 180 km of a journey at 90 km/h, then the next 260 km at 65 km/h. How long does the whole journey take?

7. Two trains leave stations 528 km apart at the same time, travelling towards each other. One travels at 85 km/h, the other at 91 km/h. After how many hours do they meet?

8. A car travels 550 km at an average speed of 110 km/h. Select ALL of the following that are TRUE: A) The journey takes 5 hours B) Increasing the speed to 125 km/h would make the journey take 4.4 hours C) In 3 hours the car has covered exactly half the journey D) At 110 km/h, the car covers 27.5 km every 15 minutes E) The journey would take 6.25 hours at 88 km/h

9. A cyclist rides 90 km at 30 km/h, then rides back the same route at 18 km/h. How long does the whole round trip take?

10. A car leaves at 06:55 and travels 315 km at an average speed of 90 km/h, including a 25-minute stop. At what time does it arrive? A) 10:30 B) 10:40 C) 10:50 D) 11:00

11. A ship sails 96 km at a speed of 24 km/h, then sails a further 189 km which takes 3.5 hours. What is the ship's average speed for the whole journey?

12. A runner completes a 32 km course: the first 24 km at 16 km/h, and the remaining 8 km at 10 km/h. How long does she take in total?

13. A car and a motorbike both leave the same point at the same time, travelling in the same direction. The car travels at 82 km/h and the motorbike at 104 km/h. How far apart are they after 2 hours 30 minutes? A) 45 km B) 50 km C) 55 km D) 60 km

14. A cargo ship travels 680 km. For the first 400 km it averages 80 km/h. For the whole journey to take 12 hours, at what average speed must it complete the final 280 km?

15. A train's journey has three legs: 150 km at 100 km/h, 189 km at 84 km/h, and 96 km at 64 km/h. What is the train's average speed for the whole journey, to the nearest whole km/h?

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